

Draft: Communication-Preserving Near-Field RIS Sensing via Phase Dithering

1 Motivation and Objective

This draft investigates a sensing mode where the RIS keeps its main focusing profile toward a served UE, while applying small phase dithers to extract environment information from CSI variations. The target is static-object detection (and optional localization) with minimal communication degradation.

Core question. Can repeated CSI under communication-preserving RIS micro-dithers provide enough information to detect a static object near the RIS in a near-field regime?

Main contributions. This draft makes six concrete contributions: (i) it introduces a communication-preserving sensing mode where RIS micro-dithers are applied around a UE-serving profile; (ii) it formulates a differential CSI model for static-object detection and optional localization; (iii) it derives a nuisance-aware EFIM/CRB feasibility framework with PEB interpretation; (iv) it presents a GLRT-based detection formulation with ROC-oriented evaluation; (v) it casts dither codebook design under explicit communication constraints such as SINR/BLER degradation limits; and (vi) it defines a practical evaluation protocol covering robustness to non-ideal RIS behavior and calibration drift.

2 Related Work and Novelty Position

Prior RIS codebook and ISAC studies mostly focus on one of the following: (i) communication beam training and link gain maximization, (ii) user localization support, or (iii) dual-purpose sensing/communication codebooks without explicitly preserving a fixed serving beam during sensing.

Representative literature. Recent works include learning-based RIS reflection codebooks for communication-oriented beam management, near-field ISAC codebooks for joint angle-range estimation, and near-field RIS localization studies based on FIM/PEB under practical phase-amplitude constraints.

Gap addressed in this draft. This work targets a different operating point: static-object sensing in near field while maintaining the RIS main profile for a served UE. Sensing is performed through small perturbations around the serving profile and differential CSI analysis.

Novelty dimensions.

- **Problem setup:** communication-preserving static-object sensing, not generic beam training.
- **Codebook design:** micro-dither codebooks constrained by communication-quality budgets.
- **Theory:** unified EFIM/CRB (estimation feasibility) and GLRT/ROC (detection capability).
- **Practicality:** explicit robustness analysis under RIS non-idealities and calibration drift.

3 System and Signal Model

Consider fixed BS, UE, and RIS positions over a sensing window. Let $m = 0, \dots, M - 1$ index RIS states (with $m = 0$ the baseline serving profile), $k = 1, \dots, K$ OFDM subcarriers, and $t = 1, \dots, T$ snapshots.

Let $\mathbf{\Gamma}_0 \in \mathbb{C}^N$ be the baseline RIS profile focused on UE. Dithered profiles are

$$\mathbf{\Gamma}_m = \mathbf{\Gamma}_0 \odot e^{j\boldsymbol{\delta}_m}, \quad (1)$$

where $\boldsymbol{\delta}_m$ is a small phase perturbation vector.

The received CSI sample is modeled as

$$y_{m,k,t} = \mu_{m,k}(\boldsymbol{\theta}) + n_{m,k,t}, \quad n_{m,k,t} \sim \mathcal{CN}(0, \sigma^2), \quad (2)$$

with

$$\mu_{m,k}(\boldsymbol{\theta}) = h_k^{\text{dir}} + h_{m,k}^{\text{RIS-UE}} + h_{m,k}^{\text{obj}}(\boldsymbol{\theta}_o) + h_{m,k}^{\text{nu}}(\boldsymbol{\eta}). \quad (3)$$

Here, $\boldsymbol{\theta}_o = [x_o, y_o, z_o, \alpha_o]^T$ are object parameters (position and complex reflectivity), and $\boldsymbol{\eta}$ denotes nuisance parameters (residual phase offsets, calibration drift, etc.).

4 Differential Observation for Robustness

To suppress common terms and hardware drifts, use differential CSI relative to baseline:

$$\Delta y_{m,k,t} = y_{m,k,t} - y_{0,k,t}, \quad m \geq 1. \quad (4)$$

For static BS/UE/RIS geometry, $\Delta y_{m,k,t}$ enhances object-dependent modulation induced by RIS dithers.

5 Feasibility via FIM and EFIM

Let $\boldsymbol{\theta} = [\boldsymbol{\theta}_o^T, \boldsymbol{\eta}^T]^T$. Under complex Gaussian noise, the Fisher Information Matrix is

$$\mathbf{J}(\boldsymbol{\theta}) = \sum_{m,k,t} \frac{2}{\sigma^2} \Re \left\{ \left(\frac{\partial \mu_{m,k}}{\partial \boldsymbol{\theta}} \right)^H \left(\frac{\partial \mu_{m,k}}{\partial \boldsymbol{\theta}} \right) \right\}. \quad (5)$$

Partition

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_{oo} & \mathbf{J}_{o\eta} \\ \mathbf{J}_{\eta o} & \mathbf{J}_{\eta\eta} \end{bmatrix}. \quad (6)$$

The equivalent Fisher matrix for object parameters is

$$\mathbf{J}_{\text{eq}} = \mathbf{J}_{oo} - \mathbf{J}_{o\eta} \mathbf{J}_{\eta\eta}^{-1} \mathbf{J}_{\eta o}. \quad (7)$$

Then CRB is

$$\text{cov}(\hat{\boldsymbol{\theta}}_o) \succeq \mathbf{J}_{\text{eq}}^{-1}. \quad (8)$$

If estimating 3D position, the position error bound (PEB) is

$$\text{PEB} = \sqrt{\text{tr}([\mathbf{J}_{\text{eq}}^{-1}]_{1:3,1:3})}. \quad (9)$$

Interpretation. This bound quantifies when micro-dithering is informative enough for localization and how much near-field geometry and dither design improve observability.

5.1 When object localization is identifiable

Localization is possible only if the measurement model is sufficiently informative and non-degenerate. In practice, the following conditions should hold:

- **Sufficient diversity across dithers:** the Jacobian vectors $\partial\mu_{m,k}/\partial\theta_o$ should not be collinear across m .
- **Frequency diversity:** enough subcarriers K and bandwidth are needed to separate range-sensitive effects.
- **Geometry diversity:** RIS-BS-UE-object geometry should avoid near-symmetric layouts that create ambiguous solutions.
- **Near-field modeling:** spherical-wave path terms must be used when the object is in Fresnel region.
- **Nuisance control:** phase/CFO/calibration drifts must be mitigated (or marginalized through EFIM).

If these conditions are weak, detection can still be feasible while localization error remains large.

5.2 How to report CRB results

For each candidate setup (SNR, M , K , RIS size, object distance), report:

- $\lambda_{\min}(\mathbf{J}_{\text{eq}})$ as an observability indicator;
- PEB from $\mathbf{J}_{\text{eq}}^{-1}$;
- empirical localization RMSE of the estimator;
- the ratio RMSE/PEB to quantify estimator efficiency.

This directly answers whether localization is feasible in each operating region.

6 Detection Formulation (Primary Task)

For static-object presence testing:

$$\mathcal{H}_0 : \Delta y_{m,k,t} = w_{m,k,t}, \quad (10)$$

$$\mathcal{H}_1 : \Delta y_{m,k,t} = s_{m,k}(\theta_o) + w_{m,k,t}. \quad (11)$$

A generalized likelihood ratio test (GLRT) is

$$\Lambda(\Delta\mathbf{y}) = \frac{\max_{\theta_o, \eta, \sigma^2} p(\Delta\mathbf{y} \mid \mathcal{H}_1)}{\max_{\eta, \sigma^2} p(\Delta\mathbf{y} \mid \mathcal{H}_0)} \underset{\mathcal{H}_0}{\geq} \gamma. \quad (12)$$

Report ROC curves, AUC, and P_D at fixed P_{FA} .

7 Dither Codebook Design Under Communication Constraint

Design dithers to maximize sensing information while preserving UE link:

$$\max_{\{\delta_m\}} \log \det(\mathbf{J}_{\text{eq}}(\{\delta_m\})) \quad (13)$$

$$\text{s.t. } \Delta \text{SINR}(\{\delta_m\}) \leq \epsilon, \quad (14)$$

$$|\delta_{m,n}| \leq \delta_{\max}, \quad \forall m, n. \quad (15)$$

Alternative criteria include minimizing $\text{tr}(\mathbf{J}_{\text{eq}}^{-1})$ or maximizing deflection coefficient for detection.

8 Evaluation Plan

- **Setup:** fixed BS/UE/RIS, near-field RIS model, static objects at known positions.
- **Data:** collect CSI over M dither states, K subcarriers, T repeats per state.
- **Tasks:** object detection; optional coarse object localization.
- **Metrics:** ROC/AUC/ P_D , localization RMSE vs CRB/PEB, SINR/BLER/throughput impact, sensing overhead.
- **Robustness:** cross-session and calibration-drift tests.

8.1 Simulation checklist for localization feasibility

1. **Baseline run:** no-object and object-present datasets with fixed geometry.
2. **SNR sweep:** at least 4–6 SNR points to identify threshold behavior.
3. **Dither sweep:** vary number of states M and dither amplitude $\delta_{\max}$.
4. **Bandwidth sweep:** vary K and effective bandwidth to test range sensitivity.
5. **Geometry sweep:** object position grid (distance and azimuth/elevation sectors).
6. **Nuisance sweep:** inject phase drift/CFO/calibration errors and compare FIM vs EFIM predictions.
7. **Communication impact:** report SINR, BLER, throughput drop during sensing.
8. **Bound comparison:** plot RMSE against PEB for each sweep.

8.2 Simulation methodology (implementation-oriented)

The following procedure can be implemented directly in a Monte Carlo simulator:

1. **Initialize geometry and channels:** set fixed BS/UE/RIS positions, near-field RIS model, and an object-location grid. Generate channels for direct, RIS-assisted serving, and object-scattered components.
2. **Construct RIS states:** compute a serving profile $\mathbf{\Gamma}_0$ for UE communication, then generate $M - 1$ micro-dither profiles $\{\mathbf{\Gamma}_m\}_{m=1}^{M-1}$.
3. **Generate CSI measurements:** for each hypothesis ($\mathcal{H}_0$ no object, $\mathcal{H}_1$ object present), each RIS state m , subcarrier k , and repeat t , synthesize $y_{m,k,t}$ under AWGN and selected non-idealities.
4. **Apply differential preprocessing:** compute $\Delta y_{m,k,t} = y_{m,k,t} - y_{0,k,t}$ and extract detection/localization features.
5. **Run detector and estimator:** evaluate GLRT (or baseline detector) for ROC/AUC and run localization estimator when $\mathcal{H}_1$ is true.
6. **Compute bounds and communication impact:** evaluate EFIM/PEB for each setup and report SINR/BLER/throughput changes during sensing.
7. **Aggregate over trials:** run sufficient Monte Carlo trials and summarize mean and confidence intervals.

8.3 Default parameter ranges

Unless otherwise specified, a practical initial sweep is:

- **SNR:** $[-10, -5, 0, 5, 10, 15, 20]$ dB.
- **Dither states:** $M \in \{4, 8, 16, 32\}$.
- **Dither amplitude:** $\delta_{\max} \in \{5^\circ, 10^\circ, 20^\circ, 30^\circ\}$.
- **Subcarriers:** $K \in \{32, 64, 128, 256\}$.
- **Repeats per state:** $T \in \{4, 8, 16\}$.
- **Monte Carlo trials:** at least 200 per setup (increase to 500 for final plots).

These ranges can be tightened once a feasible operating region is identified.

9 Expected Contributions

1. Communication-preserving near-field RIS sensing via micro-dithers around serving beam.
2. Unified estimation and detection feasibility analysis using EFIM/CRB and GLRT.
3. Dither codebook design with explicit communication-quality constraints.
4. Practical validation showing detection/localization gains at low communication overhead.

10 Conclusion

The proposed framework extends opportunistic RIS-ISAC to static-object sensing without abandoning UE-oriented beam focusing. The feasibility analysis (EFIM/CRB) and detection theory (GLRT/ROC) provide a rigorous path to justify and optimize the approach.